Checkpoint 7.1: quality and cost against 6.1 and 5.1

Oracle, vector, and hybrid experiment comparison. Generated 2026-10-04 00:20 EDT.

5	3	125	25	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

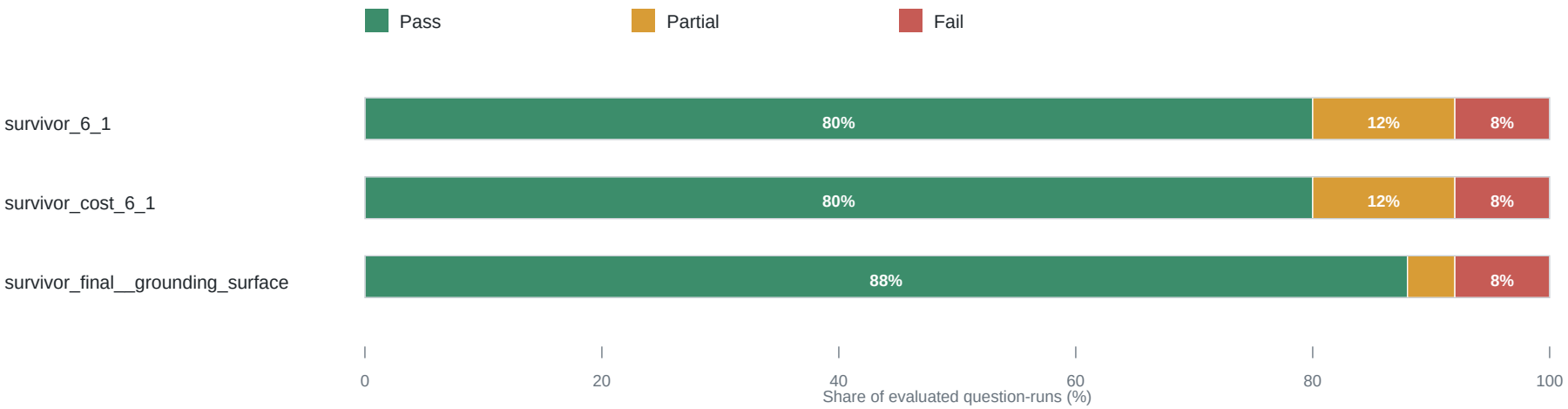
Summary

The highest observed mean quality score was 0.900 for survivor_final__grounding_surface across 1 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 100.0% for survivor_6_1. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 40.0 for Exhaustive list or time filter under survivor_final__grounding_surface. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes



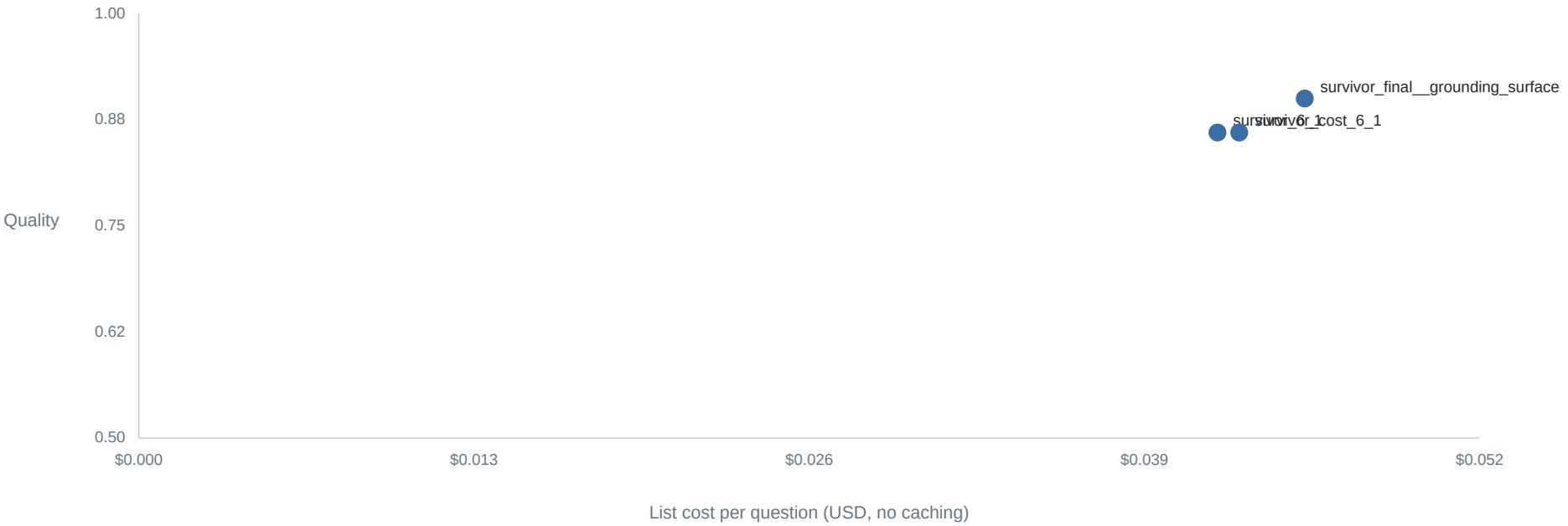
Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
survivor_6_1	3	75	80.0%	12.0%	8.0%	0.860	0.043	100.0%	100.0%	80.3k
survivor_cost_6_1	1	25	80.0%	12.0%	8.0%	0.860	0.000	100.0%	100.0%	56.6k
survivor_final__grounding_surface	1	25	88.0%	4.0%	8.0%	0.900	0.000	100.0%	100.0%	76.8k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Cost and latency

Configurations whose runs carry per-question telemetry (the Checkpoint 6.1 cost starters). Every figure, quality included, is computed over those runs only, so cost and quality come from the same question-runs.

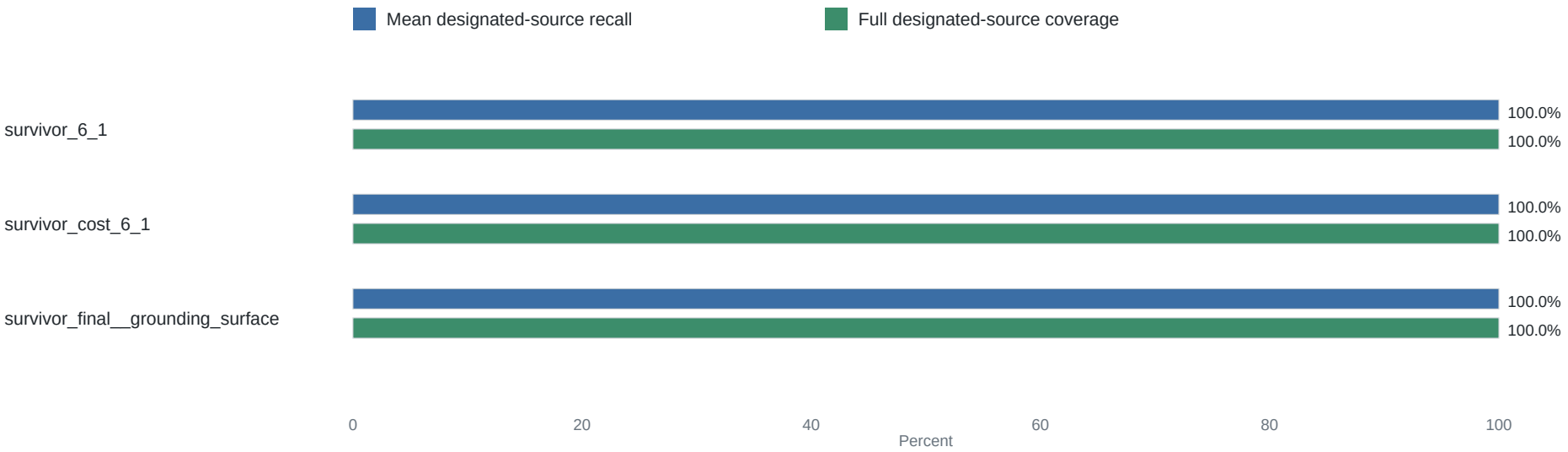
Configuration	Runs	Quality	Calls / question	Input tokens / question	Cached	List $ / question	Billed $ / question	Seconds / question	List $ per quality point
survivor_6_1	3	0.860	5.79	49.1k	38.1%	$0.0422	$0.0296	12.8	$0.0491
survivor_cost_6_1	1	0.860	6.72	49.0k	22.7%	$0.0431	$0.0356	16.1	$0.0501
survivor_final__grounding_surface	1	0.900	6.04	53.4k	22.1%	$0.0456	$0.0377	11.0	$0.0507



List cost charges every input token at the full rate; billed cost is what OpenRouter reported, with cached input at the cached rate. The judge is not included. Seconds cover the agent graph and the answer call, not the judge.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
survivor_6_1	60	0	15	0
survivor_cost_6_1	20	0	5	0
survivor_final__grounding_surface	22	0	3	0

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

	survivor_6_1	survivor_cost_6_1	survivor_final__grounding_s..
VAL01	**1.00**	**1.00**	**1.00**
VAL02	**0.17**	**0.00**	**0.00**
VAL03	**1.00**	**1.00**	**1.00**
VAL04	**1.00**	**1.00**	**1.00**
VAL05	**0.33**	**0.50**	**0.00**
VAL06	**0.33**	**0.00**	**0.50**
VAL07	**1.00**	**1.00**	**1.00**
VAL08	**0.83**	**1.00**	**1.00**
VAL09	**1.00**	**0.50**	**1.00**
VAL10	**0.67**	**0.50**	**1.00**
VAL11	**1.00**	**1.00**	**1.00**
VAL12	**1.00**	**1.00**	**1.00**
VAL13	**1.00**	**1.00**	**1.00**
VAL14	**0.83**	**1.00**	**1.00**
VAL15	**1.00**	**1.00**	**1.00**
VAL16	**1.00**	**1.00**	**1.00**
VAL17	**1.00**	**1.00**	**1.00**
VAL18	**1.00**	**1.00**	**1.00**
VAL19	**1.00**	**1.00**	**1.00**
VAL20	**1.00**	**1.00**	**1.00**
VAL21	**1.00**	**1.00**	**1.00**
VAL22	**1.00**	**1.00**	**1.00**
VAL23	**1.00**	**1.00**	**1.00**
VAL24	**0.33**	**1.00**	**1.00**
VAL25	**1.00**	**1.00**	**1.00**

Fail (0) Partial (0.5) Pass (1)

Question	survivor_6_1	survivor_cost_6_1	survivor_final__grounding_surface
VAL01	1.000	1.000	1.000
VAL02	0.167	0.000	0.000
VAL03	1.000	1.000	1.000
VAL04	1.000	1.000	1.000
VAL05	0.333	0.500	0.000
VAL06	0.333	0.000	0.500
VAL07	1.000	1.000	1.000
VAL08	0.833	1.000	1.000
VAL09	1.000	0.500	1.000
VAL10	0.667	0.500	1.000
VAL11	1.000	1.000	1.000
VAL12	1.000	1.000	1.000
VAL13	1.000	1.000	1.000
VAL14	0.833	1.000	1.000
VAL15	1.000	1.000	1.000
VAL16	1.000	1.000	1.000
VAL17	1.000	1.000	1.000
VAL18	1.000	1.000	1.000
VAL19	1.000	1.000	1.000
VAL20	1.000	1.000	1.000
VAL21	1.000	1.000	1.000
VAL22	1.000	1.000	1.000
VAL23	1.000	1.000	1.000
VAL24	0.333	1.000	1.000
VAL25	1.000	1.000	1.000

Per-question article recall

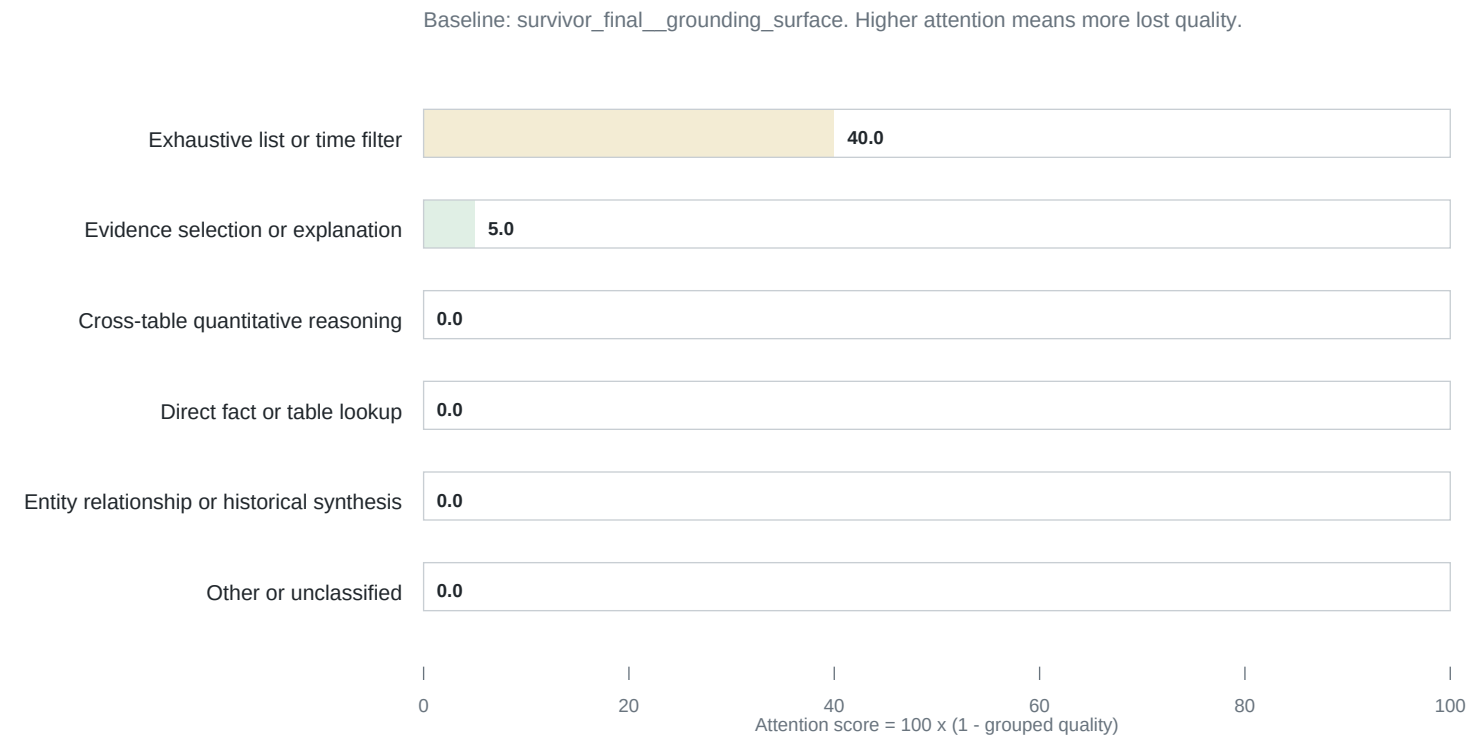
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

Question	survivor_6_1	survivor_cost_6_1	survivor_final__grounding_surface
VAL01	100.0%	100.0%	100.0%
VAL02	100.0%	100.0%	100.0%
VAL03	100.0%	100.0%	100.0%
VAL04	100.0%	100.0%	100.0%
VAL05	100.0%	100.0%	100.0%
VAL06	100.0%	100.0%	100.0%
VAL07	100.0%	100.0%	100.0%
VAL08	100.0%	100.0%	100.0%
VAL09	100.0%	100.0%	100.0%
VAL10	100.0%	100.0%	100.0%
VAL11	100.0%	100.0%	100.0%
VAL12	100.0%	100.0%	100.0%
VAL13	100.0%	100.0%	100.0%
VAL14	100.0%	100.0%	100.0%
VAL15	100.0%	100.0%	100.0%
VAL16	100.0%	100.0%	100.0%
VAL17	100.0%	100.0%	100.0%
VAL18	100.0%	100.0%	100.0%
VAL19	100.0%	100.0%	100.0%
VAL20	100.0%	100.0%	100.0%
VAL21	100.0%	100.0%	100.0%

Question	survivor_6_1	survivor_cost_6_1	survivor_final__grounding_surface
VAL22	100.0%	100.0%	100.0%
VAL23	100.0%	100.0%	100.0%
VAL24	100.0%	100.0%	100.0%
VAL25	100.0%	100.0%	100.0%

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.600	40.0	40.0%	n/a	n/a	100.0%
Evidence selection or explanation	10	0.950	5.0	0.0%	n/a	n/a	100.0%
Cross-table quantitative reasoning	1	1.000	0.0	0.0%	n/a	n/a	100.0%
Direct fact or table lookup	4	1.000	0.0	0.0%	n/a	n/a	100.0%
Entity relationship or historical synthesis	1	1.000	0.0	0.0%	n/a	n/a	100.0%
Other or unclassified	4	1.000	0.0	0.0%	n/a	n/a	100.0%

Baseline quality and failure rate use survivor_final__grounding_surface. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

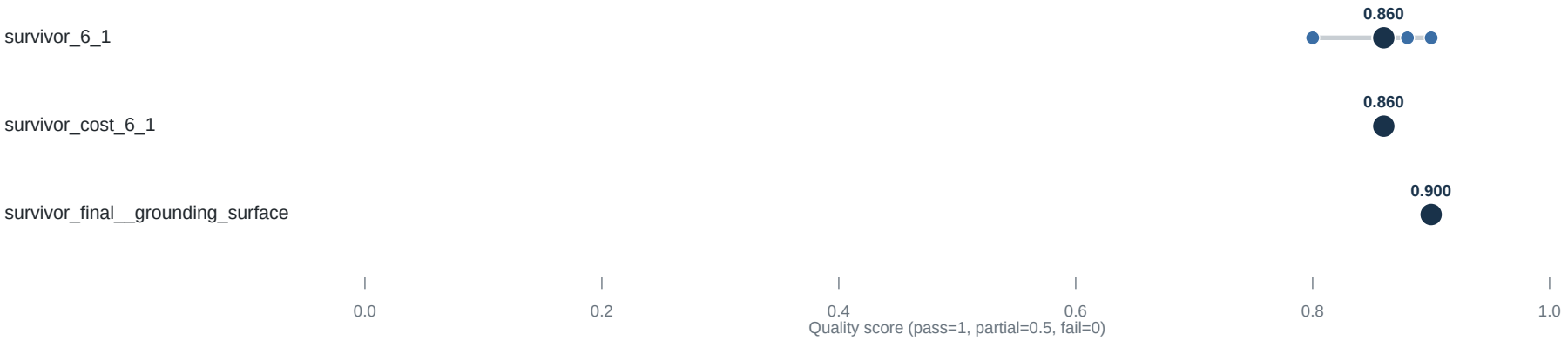
	survivor_6_1	survivor_cost_6_1	survivor_final__grounding_surface
1 article	**0.907**	**0.917**	**0.917**
2 articles	**0.833**	**0.833**	**1.000**
3+ articles	**0.167**	**0.000**	**0.000**

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	survivor_6_1	18	54	85.2%	11.1%	3.7%	0.907	100.0%
2 articles	survivor_6_1	6	18	77.8%	11.1%	11.1%	0.833	100.0%
3+ articles	survivor_6_1	1	3	0.0%	33.3%	66.7%	0.167	100.0%
1 article	survivor_cost_6_1	18	18	88.9%	5.6%	5.6%	0.917	100.0%
2 articles	survivor_cost_6_1	6	6	66.7%	33.3%	0.0%	0.833	100.0%
3+ articles	survivor_cost_6_1	1	1	0.0%	0.0%	100.0%	0.000	100.0%
1 article	survivor_final__grounding_surface	18	18	88.9%	5.6%	5.6%	0.917	100.0%
2 articles	survivor_final__grounding_surface	6	6	100.0%	0.0%	0.0%	1.000	100.0%
3+ articles	survivor_final__grounding_surface	1	1	0.0%	0.0%	100.0%	0.000	100.0%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
dazzling_minsky	survivor_6_1	25	20	4	1	0.880	100.0%	100.0%
nifty_mccarthy	survivor_6_1	25	19	2	4	0.800	100.0%	100.0%
angry_babbage	survivor_6_1	25	21	3	1	0.900	100.0%	100.0%
quirky_adleman	survivor_cost_6_1	25	20	3	2	0.860	100.0%	100.0%
awesome_valiant	survivor_final__grounding_surface	25	22	1	2	0.900	100.0%	100.0%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
angry_babbage.csv
awesome_valiant.csv
dazzling_minsky.csv
nifty_mccarthy.csv
quirky_adleman.csv

Warnings

Task-family analysis left these categories unclassified: Cross-article comparison, Diagnostic reasoning and limitations, Inference limitation, People connection: shared work and roles